

543. Diameter of Binary Tree

PLATFORM

Platform: LeetCode

DIFFICULTY

EASY

DATE

Solved: July 30, 2026

Problem Summary

Given the root of a binary tree, return the length of the diameter — the longest path between any two nodes (path doesn't need to pass through root). Length is measured in number of edges.

Approach

Post-order recursion computing height, while updating a global maxDiameter at every node. Diameter through any node = leftHeight + rightHeight.

Algo

1. Initialize maxDiameter = 0.
2. Call solve(root).
3. In solve: if root == null return 0.
4. left = solve(root.left), right = solve(root.right).
5. maxDiameter = Math.max(maxDiameter, left + right).
6. Return Math.max(left, right) + 1.
7. Return maxDiameter.

Observations

- The function does two jobs at once: computes height (returned) and updates diameter (global).
- Brute force $O(n^2)$: compute height separately at every node. Optimal $O(n)$: combine into one pass.
- solve() is named differently from maxDepth to signal it's a helper — good practice.

Time Complexity & Space Complexity

$O(n)$ — single post-order pass.

$O(h)$ — recursion stack.

Key Insights

- Diameter through any node = left arm + right arm = left + right heights.
- Diameter doesn't always pass through root — must check at **every** node.
- Global maxDiameter tracks the best answer seen so far across all nodes.
- solve() returns **height**, not diameter — height is what the parent needs; diameter is stored globally.

Problem List

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Solved

Easy

Topics

Companies

Given the **root** of a binary tree, return the length of the **diameter** of the tree.

The **diameter** of a binary tree is the **length** of the longest path between any two nodes in a tree. This path may or may not pass through the **root**.

The **length** of a path between two nodes is represented by the number of edges between them.

Example 1:

```
graph TD
    1((1)) --- 2((2))
    1 --- 3((3))
    2 --- 4((4))
    2 --- 5((5))
```

Input: root = [1,2,3,4,5]
Output: 3
Explanation: 3 is the length of the path [4,2,1,3] or [5,2,1,3].

Example 2:

Input: root = [1,2]
Output: 1

Accepted 106 / 106 testcases passed

Sahil Khan submitted at Jul 30, 2026 17:52

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Runtime: 0 ms Beats 100.00%

Memory: 46.91 MB Beats 74.24%

Code | Java

```
1 /**
2  * Definition for a binary tree node.
3  * public class TreeNode {
4  *     int val;
5  *     TreeNode left;
6  *     TreeNode right;
7  *     TreeNode() {}
8  *     TreeNode(int val) { this.val = val; }
9  * }
```

Code

Java

```
11 *     this.left = left;
12 *     this.right = right;
13 * }
14 * }
15 */
16 class Solution {
17     int maxDiameter = 0;
18     public int diameterOfBinaryTree(TreeNode root) {
19         solve(root);
20         return maxDiameter;
21     }
22
23     private int solve(TreeNode root){
24         if(root == null) return 0;
25
26         int left = solve(root.left);
27         int right = solve(root.right);
28
29         maxDiameter = Math.max(maxDiameter, left+right);
30
31         return Math.max(left, right)+1;
32     }
33 }
```

Testcase 1 Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

root = [1,2,3,4,5]

Output